Neuron mechanisms: how $d4$-dominant neurons compute degree-4 parity

`wd3_k0_N2048`, step 500k — post-nonlinearity units, uniform evaluation data

September 2026

Setup

For each neuron we analyze the post-nonlinearity activation $a = \phi(z)$, with $\phi = \text{relu}(\tanh)$ and $z = (Vh)_n$, measured immediately after the nonlinearity and before the residual addition. We call a neuron *$d4$-dominant* when, among the 15 binary-tree target modes, its largest Walsh coefficient is a four-bit parity χ_{abcd} (with $q = (x_a, x_b, x_c, x_d)$ denoting the corresponding tree quad). There are 1022 active $d4$-dominant neurons in total: 207/229/285/307 in blocks 0–3, excluding dead units. All coefficients are estimated on 2^{16} uniform samples. For each neuron, both z and a are decomposed over the complete 16-mode Walsh basis of the corresponding 4-bit cube. In both cases, this 4-bit cube explains $> 99\%$ of the variance, meaning that these neurons are, to very good approximation, functions only of those four input bits.

Mechanism: majority gates on raw bits, not composition of pairs

For the exemplar neuron (block 2, neuron 1543, quad $d4_0$), the pre-activation is approximately a *balanced sum of four singleton modes*,

$$z \;\approx\; \gamma\,(x_a - x_b - x_c - x_d), \qquad \gamma = 0.305,$$

with the four singleton terms accounting for 99.7% of $\text{Var}(z)$. In other words, before the nonlinearity there is essentially no degree-2 or degree-4 content. Since $z/\gamma \in \{-4, -2, 0, 2, 4\}$, applying $a = \phi(z)$ turns this signed linear sum into a **soft "$\geq$ 3-of-4" majority gate**: it fires most strongly, $tanh(4\gamma) = 0.84$, on the single input corner that matches the learned sign pattern in all four coordinates, and more weakly, $\tanh(2\gamma) \approx 0.56$, on the four corners that match it in exactly three coordinates (Fig. 1, left).

The resulting Walsh spectrum is almost completely reproduced by the *parameter-free* idealized gate $\phi(\gamma \sum_k s_k x_k)$ (Fig. 1, middle): roughly 70% of the variance lies in the four singleton modes, roughly 20% is distributed *equally across all six pair parities*, 8.8% lies in the four-bit parity itself, and about 1% lies in triple modes. Crucially, the neuron is not receiving already-formed upstream $d2$ features as its main inputs. The network is therefore not implementing the four-parity as an explicit $4 = 2{\cdot}2$ composition at this point; instead, this neuron acts directly on the four raw degree-1 features.

The observed coverage $\approx 1/6$, explained

Our tree-based analysis tracks only three modes on this four-bit cube: $\{\chi_{ab}, \chi_{cd}, \chi_{abcd}\}$. These are the two degree-2 parities that are actual children of the quad in the binary target tree, together with the degree-4 quad itself. The other four pair parities are real Walsh modes of the neuron, but they are *off-tree* and therefore do not count toward tree coverage. For the ideal gate, the tracked fraction is

$$\text{cov} \;=\; \frac{a_{ab}^2 + a_{cd}^2 + a_{abcd}^2}{\text{Var}(a)} \;=\; 0.151 \quad (\text{exemplar } 0.156,\ \text{population median } 0.142),$$

which matches the observed value of about 0.16 (Fig. 2, middle). Thus the apparent "1/6" coverage is not the result of the neuron's variance being symmetrically divided into six equal pieces. It is simply the fraction of the majority-gate spectrum that happens to lie on the three modes we track in the target tree. The remaining $\sim$84% is mostly singleton power ($\sim$ 70%) plus the four off-tree cross-pair modes $\chi_{ac}, \chi_{ad}, \chi_{bc}, \chi_{bd}$ ($\sim$ 14% in total). Each of these cross pairs is about as strong as either of the two tree pairs; they are excluded only because they are not nodes of the chosen binary tree.

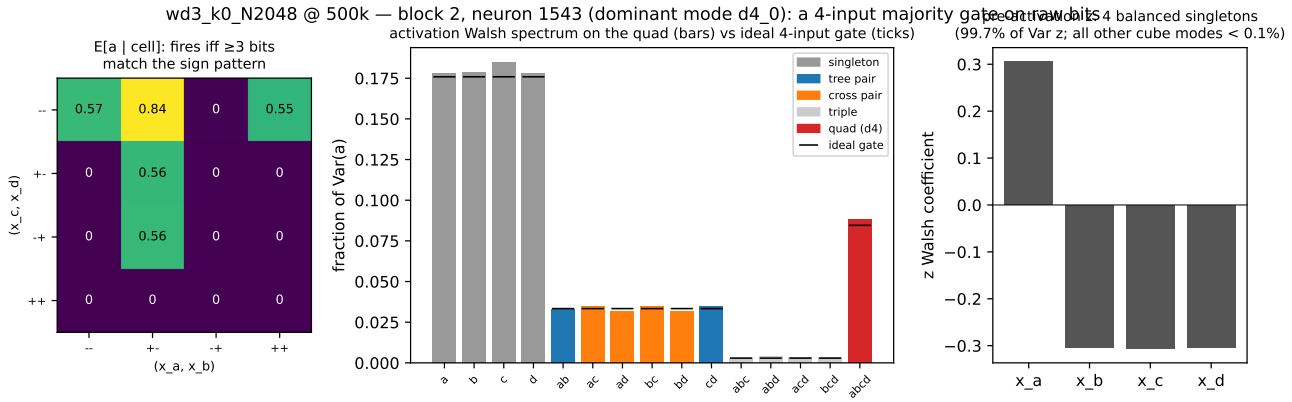


Figure 1: Exemplar $d4$-dominant neuron (block 2, #1543). Left: conditional means over the 16 cells of its quad — a $\geq$ 3-of-4 majority gate. Middle: measured Walsh spectrum (bars) vs. the ideal gate prediction (black ticks); colors mark the tree-tracked modes. Right: the pre-activation is four balanced singletons.

Population: universal gates, plus quad-amplifiers in the last block

This mechanism is not specific to the exemplar. Among the highest-ranked $d4$ neurons in blocks 0–2, **100%** satisfy the balanced-singleton input criterion ($\geq 90\%$ of Var(z) in the four singleton modes, with balance ≥ 0.7); across all active $d4$-dominant neurons, the fraction is 72.6% (Fig. 2, left). The population contains all three possible sign-pattern classes, 4:0/3:1/2:2 = 145/646/376. These correspond to majority/corner detectors centered on different regions of the four-bit cube rather than to one unique gate.

The main exception appears in the final residual block. There are 160 neurons—155 of them in block 3—with tree coverage close to 1. Their pre-activations are no longer dominated by raw singleton features. Instead, the already-formed degree-4 parity is itself the dominant input component: in z, the median variance allocation is 84% to χ_{abcd}, 14% to the two tree-pair modes, and only 0.5% to the singletons. These neurons are therefore better described as **pass-through amplifiers**: blocks 0–2 create degree-4 parity through majority-gate neurons, while their large singleton and off-tree pair components can cancel when many neurons are combined; block 3 then receives the cleaner population-level parity feature and re-amplifies it.

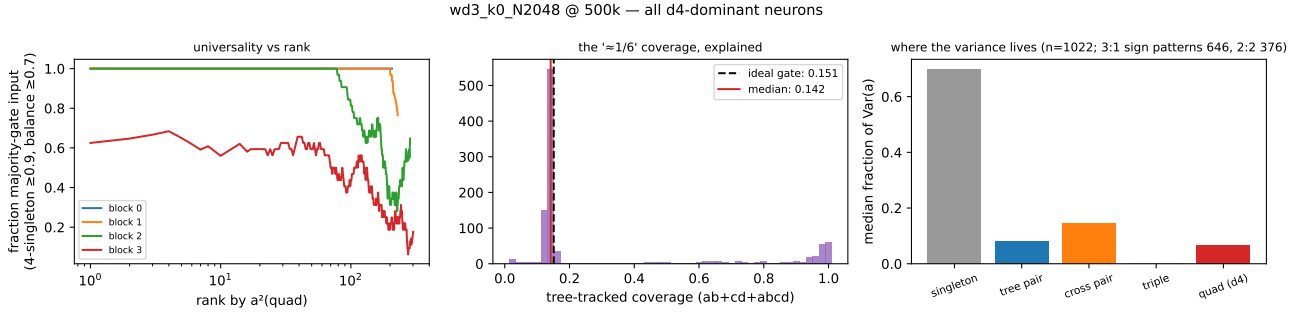


Figure 2: All active $d4$-dominant neurons. Left: fraction with majority-gate input vs. rank by quad power (blocks 0–2 at 1.0; block 3 lower — see amplifiers). Middle: tree-tracked coverage concentrates at the ideal-gate value 0.15 ($\approx$ the observed 1/6), with the amplifier subpopulation at ≈ 1. Right: median variance allocation — most of a $d4$ neuron is off-tree.

Takeaway. In `wd3_k0_N2048`, degree-4 parity is not primarily computed by composing the network's existing $d2$ features. Instead, most $d4$-dominant neurons in blocks 0–2 implement soft majority/corner detectors directly on the four underlying raw bits. Such a gate naturally contains the desired degree-4 parity, but also much larger singleton components and equal-strength degree-2 components on all six pairs, including four pairs that are not part of the target binary tree. This is why the tree modes explain only about 1/6 of the variance of a typical $d4$-dominant neuron. The clean degree-4 parity is therefore best viewed as a *population-level* feature: unwanted

singleton and off-tree components cancel across neurons, while the shared parity component survives. In the final block, dedicated amplifier neurons then receive this consolidated parity and carry it much more directly.

Artifacts: `gdrive:parity/analysis_output/neuron_mechanisms/wd3_k0_N2048/step_00500000/` (figures, `d4_per_neuron.csv`, exemplar tensors, this report; d2 corner-unit probe in the same folder).